# When Locality Helps or Hurts: A Trace Study of Graph-Based Nearest-Neighbour Search

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Computer Science / Computer Architecture (HPC)
Companion paper to the thesis, 10 November 2026

Abstract—Storing co-accessed vectors together is a natural optimisation for graph search, but row reuse competes with bank parallelism. This study records stock HNSW demands on SIFT100k and measures their completed service in a pinned HBM2 simulator. It compares simple and trained placements, including published graph-reordering routines, while preserving search work and recall. Memory phases withhold later graph expansions and expose only available candidate reads to request reordering. On ten reference query splits, round-robin on BFS changes cycles by -14.58% at 512 bytes and +3.43% at a hypothetical 128-byte footprint under default settings. Outstanding limits and address mapping can reverse the recommendation at the same footprint. A 50-query training profile selects an order per configuration and freezes it for held-out service, matching the retrospectively best fixed order in all 140 tested cases. The original contribution is the measured failure cases and conditional alternatives, their command-level explanation, and explicit service-cost limits for a proposed near-memory fetch interface. The artifact makes these choices reproducible at fixed search work. Profiling remains exploratory; computation, caches, real overhead and integrated PIM execution remain outside the measured model.

Index Terms—approximate nearest neighbours, graph search, memory layout, DRAM simulation, processing-in-memory

## I. Domain and motivation

The memory wall describes a mismatch between the processor's ability to compute and the memory system's ability to supply data [3]. Latency is the delay before a read completes; bandwidth is the amount transferred per unit time. Pointer-following may expose too little independent work to use peak bandwidth. Processing-in-memory (PIM) moves computation in or near memory to reduce some transfers, but still faces data dependencies, DRAM timing and implementation costs.

### A. Graph-based vector search

A vector-search system stores an array of numbers for each item and retrieves items close to a query array under a distance function. Image retrieval and embedding search are examples. An exact scan compares the query with every stored vector. Approximate nearest-neighbour (ANN) methods reduce that work by examining promising candidates, accepting that some exact neighbours may be missed.

Hierarchical Navigable Small World (HNSW) search represents vectors as nodes in a proximity graph [1]. To expand a node, it reads an adjacency list of neighbour IDs, fetches candidate vectors, and evaluates their distances. Those distances guide the next expansion. For example, expanding node 17 might reveal candidates 4 and 29. Their addresses become useful through the earlier graph work; the complete future request sequence is not initially known to a real execution.

### B. Locality and parallelism

A DRAM bank has a row buffer holding an open row. An activate command (ACT) opens a row, a read command (RD) transfers data from it, and a precharge command (PRE) closes it before another row can open. A request to the already open row is a row hit. Avoiding repeated row changes can save commands and waiting time.

Banks also provide parallel service opportunities. While one bank waits for a timing constraint, another may make progress. Consecutive requests concentrated in one bank can reduce these opportunities. A row hit therefore need not complete immediately: command spacing, shared resources and other requests still matter.

The controller's choices depend on concurrency. With one outstanding read—a request accepted but not completed—there is little opportunity to overlap different reads. With many outstanding reads, bank diversity can be useful. The page policy controls when rows close, the address mapper selects a bank and row from address bits, and the scheduler chooses among queued commands. These are experimentally controlled variables here.

A CPU cache adds another layer: a cache hit can avoid a DRAM request entirely. A DRAM row hit still requires memory service. Keeping that distinction clear is essential when comparing this study with cache-oriented graph reordering.

## II. Prior work and the remaining question

Gorder's structural graph ordering [10] and the profile-guided Porder in Coleman et al.'s NeurIPS 2022 paper [8] are measured placement baselines here. The latter's cache-aware query-time improvements, up to 40% in its evaluation, address a different execution layer. This study

adds their relative DRAM-service effects with search work fixed and availability explicit; it does not reproduce their application result. DiskANN further illustrates complete storage-system design [11].

PIM research combines placement with computation and communication [3]. DRIM-ANN includes arithmetic lookup tables, load balancing and I/O optimisation on computationally enhanced PIM [12]. UpANNS evaluates commercial-PIM resource management and ANN co-design [13]; PIMANN studies persistent kernels and fine-grained scheduling [14]. PIMCQG, whose public revision states acceptance at SC26, directly addresses graph ANN on PIM with a compact index and asynchronous execution [15]. These are literature comparisons, not benchmarked implementations in this artifact. It supplies isolated candidate-fetch comparisons and cost constraints within that broader design problem. Their service percentages cannot rank this study against those systems' application speedups. This targeted comparison makes no exhaustive priority claim for the general locality–parallelism interaction.

## III. Research questions and contribution

A layout can store related nodes close together. This may improve memory service, but "close" is only an address property. The memory controller decides which commands can issue and which requests must wait. We ask three connected questions. RQ1: how does request order interact with concurrency? RQ2: which tested placements and available-request schedules improve service at fixed search work, and can training select an order? RQ3: which conclusions survive the selected footprint, availability, controller and graph changes?

The study measures DRAM stream service: controller cycles to complete an explicitly supplied read sequence. Computation, caches, scratch structures, mapping lookups and instruction-level control overhead are excluded. Flat replay offers a known trace; bounded replay adds memory boundaries between expansions. Processing-in-memory (PIM) motivates the study but is not implemented as an application platform.

The core empirical contributions are measured limits of locality-oriented placement (C1) and conditional scheduling gains with command evidence for reversals (C2). They support an exploratory execution recommendation with cost thresholds (C3), using established selection and cost models. Designers can test a fetch order under the intended availability and mapping assumptions, then compare its saving with integration costs. Graph reordering, PIM-based ANN, locality optimisation and fine-grained scheduling are established; the original outputs are the evidence and its interpretation for this workload.

## IV. Study design and methodology

### A. What stays fixed and what changes

Figure 1 shows the experiment chain. Observer hooks in stock hnswlib record each required vector read and each expanded adjacency list, including upper graph layers and the entry vector. Search decisions and result ordering remain in stock code. The driver verifies every observed result against unobserved search, and tests compare 800 results with independently built indexes, including ties, narrow search settings and shuffled insertion.

A layout assigns every corpus node a unique vector slot. If node $v$ is assigned slot $\pi(v)$, its vector begins at $A_V + \pi(v)b$, where $A_V$ is the array base and $b$ is the stored vector size. All layouts for one graph and split receive the same logical accesses and return the same answers. They change addresses, rather than the number of distance evaluations.

Adjacency lists occupy separate, aligned allocations independent of vector layout. A list containing $d$ neighbour IDs requires $4 + 4d$ bytes: a four-byte header plus four bytes per ID. The emitter covers each payload with the simulator's native 32-byte reads. Thus a 512-byte vector needs sixteen reads and a 128-byte vector needs four. No cache or duplicate-read coalescing is assumed.

### B. The initial five layouts

Identity keeps corpus order; random supplies a shuffled control. Breadth-first search (BFS) placement visits graph neighbours before the next frontier and stores that visitation order. Community placement groups nodes using repeated neighbour-label updates. These two methods use graph structure without training queries.

The query-aware method learns which different vectors are accessed close together in training searches. It counts pairs using a bounded history of 16 distinct nodes within each query. It then greedily places heavily co-visited nodes together, without crossing the current grouping boundary. This can capture shared visits between nodes that have no direct graph edge. It optimises a co-visitation objective, however, rather than simulator cycles.

The verified contiguous same-bank/row span is 1,024 bytes under the evaluated mappings. A fresh aligned block fits two 512-byte vectors or eight 128-byte vectors, so the heuristic is rebuilt for each footprint. The smaller footprint is hypothetical repacking: float32 search decisions, distances and recall remain unchanged. It is not an implementation of quantised HNSW or product quantisation (PQ), which changes representation and distance evaluation [2].

### C. Memory model and workload

The Ramulator 2.1 implementation is pinned to revision 7bead6cb7c4b [4], [5]. We configure one HBM2 channel using the HBM2_8Gb and HBM2_2000Mbps presets. The GenericDDR controller selects at most one command per controller cycle. Default settings are an open-page policy, readiness-first FRFCFS scheduling, 64 outstanding reads,

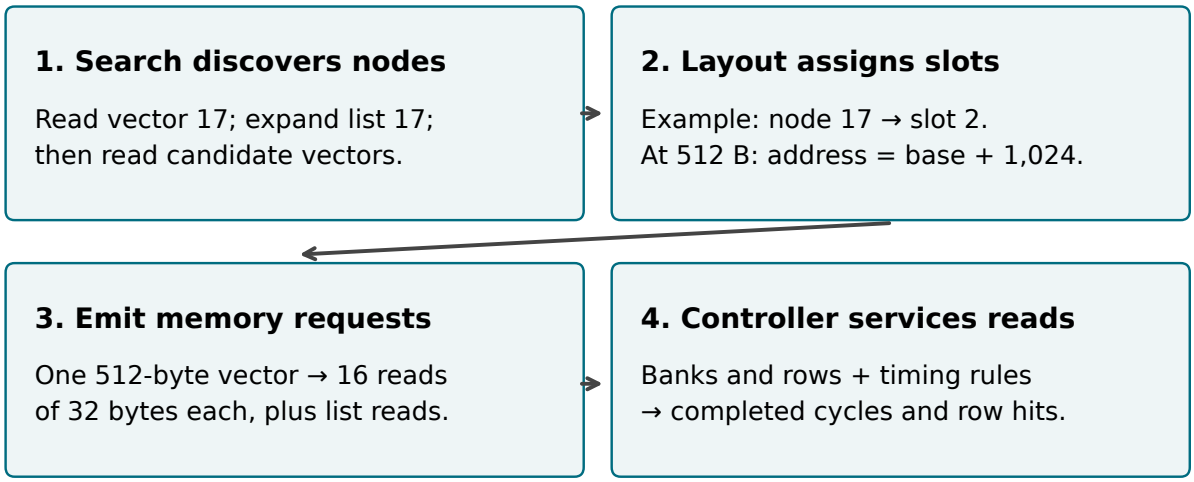



Fig. 1. A logical search determines which data are needed. A layout determines their addresses. The emitter turns payloads into native requests, and the controller determines service time. The node and slot numbers illustrate the transformation.

a 32-entry read buffer and one issue opportunity per cycle. The bound counts transactions: 64 native reads represent four full 512-byte vectors or sixteen 128-byte vectors before adjacency traffic. Footprint therefore changes both grouping capacity and the amount of logical vector work inside the bound. The full thesis documents address-bit assignments and timing assumptions; integration tests verify the actual mapper and native transaction size.

Each query drains its reads before the next query issues, while open-row state is retained. The frontend waits for every completion. Every replay must reconcile trace reads, submissions, completions, served requests and row classifications. Optional command observers preserve the baseline timing and record the commands used in the mechanism analysis.

The primary data are the first 100,000 SIFT vectors from the ANN-Benchmarks mirror [7], with 128-dimensional squared Euclidean distance. HNSW uses $M = 16$, construction search width 200, query search width 64 and $k = 10$. These parameters control connectivity, construction effort, search effort and the number of requested answers, respectively. Exact top-ten neighbours are recomputed over this same corpus subset. Returning nine of those ten would give Recall@10 of 90%.

### D. Repetition and interpretation of intervals

Each of ten seeded shuffles of the 10,000-query pool supplies 500 training and 200 held-out test queries. They are disjoint within a split. The same test queries evaluate every layout, and graph seed 100 and corpus insertion order are fixed in this primary comparison.

For split $s$ and layout $L$, the paired cycle difference is

$$d_{s,L} = 100\,(T_{s,L}/T_{s,\mathrm{BFS}} - 1)\,. \tag{1}$$

Negative means fewer cycles than BFS. For example, 99,700 against 100,000 cycles is $-0.3\%$. Pairing removes the confounding effect of testing one layout on more difficult queries. We report the mean difference and a Student 95% interval over the ten split effects, using nine degrees of freedom [16]. It describes uncertainty in this mean under the fixed graph and model, not the spread of individual query latencies or all possible datasets.

Three controller settings are repeated across all ten splits. A separate construction study evaluates graph seeds 101 and 102 in corpus order, and graph seed 100 with insertion-order seed 17, each on splits one through three. External corpus IDs are preserved after shuffling. Means and observed ranges for those selected constructions are reported separately; they are not pooled into a population confidence interval. All comparisons remain exploratory, without a family-wise multiplicity claim.

## V. Results: from the simple case to graph search

### A. A controlled reversal and its mechanism

The sequential and permuted microbenchmarks read the same 4,096 vectors, each 128 bytes, for 16,384 transactions. At the default concurrency, sequential input has 96.78% classified row hits but takes 30894 cycles; permuted input has 76.85% hits and takes 28968 cycles. More hits coincide with more time.

Table I separates cycles that issue a command from cycles that do not. Permutation requires more row operations, but at high concurrency it more than compensates through fewer no-command cycles. Its offered stream spans an average of 14.81 distinct bank instances per 64 consecutive requests, compared with 2.97 for sequential order. This is input diversity, not measured queue occupancy. The recorded read-command timeline in the full thesis shows the different bank exposure during service.

TABLE I
Command-level explanation of the controlled ordering experiment. Limit is the maximum number of outstanding reads. ACT opens a row; PRE closes one. "No command" counts controller cycles in which no command issues, including timing waits and final draining. It is not a count of one specific stall cause.

| Limit | Order | Cycles | ACT | PRE | No command |
|---|---|---|---|---|---|
| 64 | Sequential | 30894 | 528 | 38 | 13930 |
| 64 | Permuted | 28968 | 3793 | 3319 | 5458 |
| 1 | Sequential | 325786 | 588 | 166 | 308482 |
| 1 | Permuted | 398592 | 4044 | 1135 | 376825 |

With the outstanding limit reduced to one, the result reverses: avoiding row changes becomes more valuable because other reads cannot hide the waiting. These controlled interventions and command records explain a locality–parallelism trade-off without attributing every idle cycle to one timing constraint.

Additional experiments vary disjoint request-sorting windows, contiguous run length, alignment and placement perturbations. Sorting the entire known stream exactly reconstructs sequential input, but dependent graph search may not know those future addresses in advance. The run experiments also distinguish the verified 1-KiB address span from an empirical saturation point, which changes with alignment and the tested grid. Full tables are retained in the thesis and artifact.

## B. Placement effects on held-out SIFT queries

Mean Recall@10 is 98.99%, with split-level interval [98.77, 99.22]%. Table II compares layouts at this fixed search operating point. The query-aware layout saves a small fraction of cycles at 512 bytes and costs a small fraction at 128 bytes. The sign is a measured result of each footprint-specific layout and controller interaction.

TABLE II
Flat-replay mean cycle difference relative to BFS and paired 95% interval over ten query splits. Negative is faster. The 128-byte case changes storage footprint while retaining float32 search.

| Layout | 512 bytes (%) | 128 bytes (%) |
|---|---|---|
| Identity | +0.30 [+0.24, +0.36] | +2.68 [+2.60, +2.76] |
| Random | +0.31 [+0.12, +0.49] | +2.92 [+2.80, +3.04] |
| BFS | +0.00 [+0.00, +0.00] | +0.00 [+0.00, +0.00] |
| Community | +0.15 [+0.01, +0.29] | +2.35 [+2.17, +2.53] |
| Query-aware | -0.31 [-0.37, -0.26] | +0.17 [+0.11, +0.24] |

At 128 bytes on split one, query-aware placement improves the row-hit fraction yet takes 2,848 more cycles. Its command record contains 12,516 fewer activations and saves 25,125 command-issue cycles, while adding 27,973 no-command cycles. The reversal remains when adjacency requests are removed. Adjacency-only streams are identical for the two layouts and produce identical timing. These tests locate the effect within vector service and scheduling interactions; component times cannot be added because mixing traffic changes the schedule.

The distinction between locality within and between vectors also matters. Under an ideal uninterrupted sequence with a first-read miss, the remaining fifteen of sixteen reads of a 512-byte vector could hit the open row: 93.75%. For four reads the analogous fraction is 75%. These are illustrative within-vector opportunities, not predictions of the simulator's hit count. A high aggregate hit rate can therefore coexist with little useful placement of different vectors.

## C. How robust is the ranking?

The repeated settings change concurrency, row closure and bank mapping separately. Table III reports the query-aware/BFS differences; all five layouts are retained in the artifact. The query-aware/BFS sign is consistent across all ten splits in each repeated setting: faster at both widths with one outstanding read or the alternate mapper, and slower under cap four. Under alternate mapping, random placement is faster still at 512 bytes. These estimates extend the earlier one-split observations. Because the settings were chosen after inspecting that exploratory matrix and include its first split, this is a robustness extension rather than an independent confirmation.

TABLE III
Selected controller settings repeated across ten paired splits. Other controller and heuristic ablations remain single-split exploratory results.

| Setting | Bytes | Δ QA/BFS (%) | 95% interval |
|---|---|---|---|
| Outstanding: 1 | 512 | -0.13 | [-0.14, -0.13] |
| Outstanding: 1 | 128 | -1.26 | [-1.30, -1.21] |
| ClosedCAP, cap 4 | 512 | +0.89 | [+0.79, +0.99] |
| ClosedCAP, cap 4 | 128 | +5.27 | [+5.15, +5.39] |
| Alternate mapper | 512 | -4.02 | [-4.28, -3.75] |
| Alternate mapper | 128 | -4.12 | [-4.26, -3.99] |

The additional graph constructions are compared on matching query splits, with full ranges retained in the thesis. Under flat replay at 512 bytes, query-aware is faster than BFS in every tested construction/split pair. At 128 bytes, the original graph has a three-split mean of +0.119%, compared with -0.191% and -0.705% for graph seeds 101 and 102. Shuffled insertion gives splits on both sides of zero. The original 128-byte loss therefore does not generalise even across these graph constructions. Changing the graph can also change the search work and recall, so the cycle ratios always compare layouts within that graph. Recall across the evaluated graph/split conditions ranges from 98.55% to 99.35%.

Training-size, history-length and grouping-size ablations expose the heuristic's dependence on its training objective. They are not an exhaustive tuning search. Default training covers 97.7% of test vector accesses by node occurrence. High node coverage alone does not show that useful pair relationships or controller opportunities were learned.

D. Stronger baselines and available requests

1) Follow-up design: The initial heuristic alone cannot establish whether profile-guided placement is useful. We therefore add the published RCM, Gorder and Porder routines of Coleman et al. [8]. RCM reverses a degree-ordered Cuthill–McKee traversal [9]. Gorder [10] scores graph relationships to a recent placement window; Porder weights them using query traversal frequencies. Both use the supplied five-node window. Porder assigns one plus the training traversal count to each base-layer edge. The local query-aware method instead weights co-visited pairs and uses footprint-specific block grouping.

The adapter applies the supplied permutations to the separate vector array and preserves the source's window-expiration condition. An ordered-set helper replaces the weighted linked-list queue with matching score and tie semantics. All three methods match the untouched reference on 30 weighted graphs, and sanitizer runs preserve the normal mappings on the full graph. This evaluates their placement routines under a different memory representation; it does not reproduce the authors' cache-aware application benchmark.

To test available work, the follow-up drains an adjacency payload before offering its candidate vectors, then drains those vectors before the next expansion. Entry reads receive separate phases. Candidate membership is checked against the preceding list. Figure 2 illustrates this memory-boundary model. It permits overlap among a phase's candidates under the same 64-transaction bound, without executing distance computation. A phase-to-query map reconstructs original-query service including inter-phase gaps.

Address sorting and transaction round-robin use only a phase's available reads. The former orders their addresses; the latter sends the first chunk of every vector, then the second, and so on. Every duplicate transaction remains present, and none crosses a phase. Scheduling and buffering overhead are omitted. This is a defined dependency sensitivity, not instruction-level HNSW or an integrated PIM engine.

The fixed matrix adds 494 replays on all 19 graph/split pairs and both widths: three new flat placements, six bounded placements, and two schedules on BFS and query-aware placement. Four further command-recording repeats diagnose the pilot's scheduling contrast without changing its timings. All 791 earlier results remain available. These extensions reuse previously evaluated queries and remain exploratory.

2) What placement and ordering change: Table IV compares published layouts against BFS with matching expansion boundaries. The full thesis retains all six placements against identity and BFS, under both flat and bounded replay. This prevents a result from the local heuristic from standing in for the whole class of trained layouts.

TABLE IV
Published placement routines under bounded replay. Mean cycle differences and 95% intervals over ten reference splits. Negative means fewer cycles than matching BFS.

| Bytes | Layout | Δ vs. BFS (%) |
|---|---|---|
| 512 | RCM | -0.16 [-0.21, -0.11] |
| 512 | Gorder | -0.20 [-0.24, -0.16] |
| 512 | Porder | -0.38 [-0.47, -0.28] |
| 128 | RCM | -0.47 [-0.55, -0.38] |
| 128 | Gorder | -1.33 [-1.42, -1.25] |
| 128 | Porder | -1.19 [-1.33, -1.04] |

Figure 3 shows the size and sign of the scheduling effects. Table V reports comparisons that change only order within matching BFS phases. At 512 bytes, the round-robin mean saving is 5,802 controller cycles per original query. Across the 19 graph/split cases, 100.0% of query records improve in each case. The empirical 95th-percentile service differences range from -15.24 to -13.61%. These are descriptive within-case checks, not independent replications or application latency guarantees. The round-robin sign persists across the selected constructions on both evaluated placements. Placement itself is less uniform: expansion boundaries reverse a small query-aware 512-byte advantage on graph seed 101, split two. Construction-specific ranges and all scheduling results are retained in the thesis and artifact.

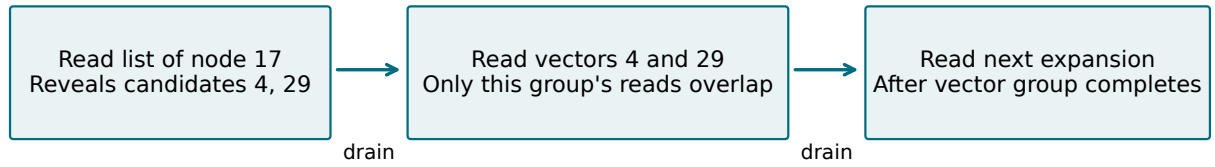
TABLE V
Available-request schedules on BFS. Mean cycle differences and 95% intervals over ten reference splits, relative to original-order bounded replay. The smaller footprint remains hypothetical.

| Bytes | Schedule | Δ vs. original (%) |
|---|---|---|
| 512 | Sorted | +0.19 [+0.13, +0.24] |
| 512 | Round-robin | -14.58 [-14.64, -14.51] |
| 128 | Sorted | +0.17 [+0.12, +0.21] |
| 128 | Round-robin | +3.43 [+3.31, +3.55] |

The first split's command accounting explains the contrasting widths. At 512 bytes, round-robin adds 277,894 command cycles but removes 1,429,356 no-command cycles, saving 1,151,462 overall. Offered bank diversity within complete 64-request phase windows grows from 4.75 to 15.07, despite a lower row-hit fraction. At 128 bytes, extra row operations and no-command cycles together cost 103,435 cycles. Its original stream already exposes more vectors through the transaction bound. The accounting supports the role of bank exposure and row reuse; it does not isolate every timing restriction. Even mean read latency falls in both cases, showing why completed-work timing remains necessary.

E. From a conditional result to an execution choice

1) Scheduling sensitivity: We add 360 BFS replays over ten reference splits and both footprints, changing one




**Example with two transactions per vector:**

Original: 4a 4b 29a 29b

Round-robin: 4a 29a 4b 29b

Address sorting orders this same group.

No read moves across a drain boundary.

Fig. 2. Expansion-bounded replay. Only transactions within the candidate-vector group are reordered. The illustration uses two chunks per vector; the experiment uses sixteen at 512 bytes and four at 128 bytes. Drain boundaries wait for completions while retaining row state.

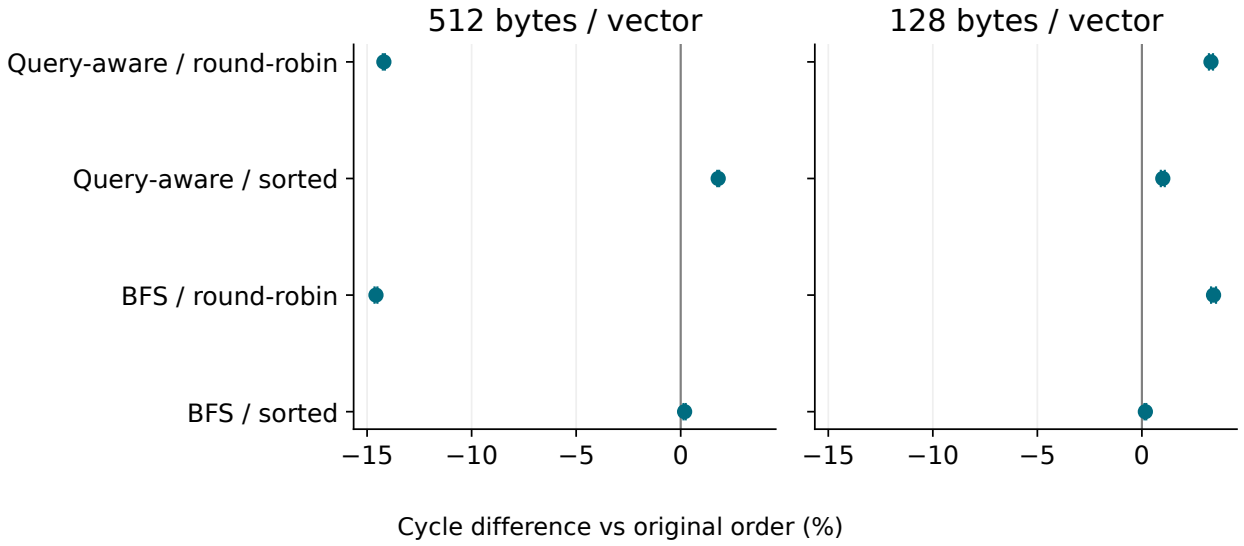



Fig. 3. Available-request scheduling on the reference graph, ten paired splits. Points and bars show means and 95% intervals. Both panels use the same scale; negative differences favour the schedule.

setting at a time: outstanding limits 16, 32 and 128; consecutive candidate batches of at most two or eight, each drained before the next; and the alternative mapper ChRaBaRoCo. The default is reused. A transaction limit bounds uncompleted reads; a batch bounds vectors exposed before a drain. These are separate interventions, not a full factorial study. Hashes and phase maps verify unchanged reads, candidate membership and original-query accounting.

The result prevents a footprint-only rule. At 128 bytes, round-robin changes cycles by -10.61% with 16 outstanding reads, compared with +3.43% at the default. Under the alternative mapper it changes cycles by +112.71% at 512 bytes and +42.59% at 128 bytes. Sorting's alternative-mapper 128-byte difference is -0.34%. The full thesis retains every fixed-order effect, paired interval and query-level diagnostic.

2) Training-selected vector fetch: The optional execution path profiles original, sorted and round-robin fetch on the first 50 training queries, chooses minimum total service cycles and freezes that choice for all 200 held-out queries. Exact ties prefer original, then sorted, then round-robin. The configuration includes placement, footprint, batch limit and memory settings. This adds 420 calibration replays and 140 choices; held-out evaluation reuses the exact matching measured stream. Within each split, calibration and held-out IDs are disjoint. The extension was motivated by already inspected results, so its evaluation remains

TABLE VI
Training-selected BFS schedule versus matching original order. Mean cycle difference (%) and paired 95% interval over ten reference splits. Negative means fewer cycles. The 128-byte case is hypothetical repacking. All six sensitivity conditions are retained. Zeros denote selection of the original-order baseline.

| Condition | 512 bytes | 128 bytes |
|---|---|---|
| Default 64 | -14.58 [-14.64, -14.51] | +0.00 [+0.00, +0.00] |
| 16 outstanding | -33.08 [-33.21, -32.94] | -10.61 [-10.68, -10.54] |
| 32 outstanding | -21.86 [-21.95, -21.78] | +0.00 [+0.00, +0.00] |
| 128 outstanding | -14.58 [-14.64, -14.51] | +0.00 [+0.00, +0.00] |
| Batch 2 | -7.77 [-7.79, -7.76] | -1.48 [-1.51, -1.46] |
| Batch 8 | -15.55 [-15.60, -15.50] | -2.48 [-2.52, -2.43] |
| Alt. mapper | +0.00 [+0.00, +0.00] | -0.34 [-0.36, -0.33] |

exploratory despite training-only selection.

Table VI reports held-out service. The selector chooses original/sorted/round-robin 40/10/90 times. It matches the best of those three fixed orders in all 140 cases: seven settings, two footprints and ten splits on one graph/query pool. Each setting receives the same choice across its ten splits. These shared-workload cases are not independent trials estimating a universal success probability. At default 512 bytes, the selected BFS path differs from published Porder with original order by -14.25 [-14.36, -14.14]%; at 128 bytes it differs from Gorder by +1.35 [+1.26, +1.44]%. All fixed-order comparisons remain in the thesis; published placement plus selection remains untested.

## VI. What we learned and practical implications

Statistical, practical and architectural significance answer different questions. The flat-replay query-aware/BFS interval at 512 bytes excludes zero, but its saving is only about 97.3 simulated cycles per query. That is weak standalone justification for a trained layout. Default bounded BFS round-robin saves 5,802 cycles per query at 512 bytes, while at 128 bytes its saved-cycle count is -541, a regression. The larger contrast motivates testing an implementation; the reversals explain why row hits or footprint alone cannot select its schedule.

In a simplified additive model, a fractional service reduction $r$ reduces total time by $fr - h$, where $f$ is the baseline fraction spent in that service and $h$ is extra overhead as a fraction of baseline time. Neither quantity is measured here, and overlapping compute with memory can invalidate this model. No application benefit is predicted. Profiling, buffering, mapping, dispatch and host/PIM transfers must be charged on a common target; CPU seconds cannot be divided by simulated controller cycles to infer deployment amortisation.

For two schedules on the same accelerator, charge their additional dispatch, buffering, synchronisation and transfer costs; shared work is not an extra scheduling cost. Consuming the default 512-byte saving of approximately 5,802 controller-equivalent cycles per query erases the gain. Calibration also averages 5.72 million service cycles there. Dividing it by the mean per-query saving gives about 985 query services before overhead, or about 1,971 if extra work consumes half the saving. These are serial service-only illustrations from retained means, not deployment break-even predictions.

The two stored 64-bit mapping directions require 1.53 MiB for 100,000 nodes, excluding container overhead. A deployed implementation may use a different representation, but its translation, storage movement and rebuilding costs need measurement. Shared-host preprocessing timings are retained for accounting; they are not isolated platform benchmarks.

The practical evaluation sequence is: hold answer quality and logical work fixed; measure completed time as well as locality; check the mapper and realistic concurrency; and compare the gain with the costs of integrating the layout. Simple graph placement is a useful baseline. The best measured layout under one controller is a candidate for further evaluation, not a universal recommendation.

A fixed order already chosen correctly for a configuration gives the same replay benefit without repeating calibration. Profiling therefore helps choose a schedule; it does not outperform that correctly chosen fixed schedule. Its target cost must be recovered before configuration or workload changes invalidate the choice. No reprofiling interval is established. A deployment should monitor mean and tail behaviour, compare against its validated fixed baseline after relevant changes, and abandon a choice that regresses after overhead. These safeguards are proposed, not evaluated components of the selector.

A concrete PIM integration would keep traversal, visited state and queues on the host, while an engine gathers resident vectors and computes distances. The host sends the query once and supplies batches of resolved addresses and order tags after reading neighbour lists. The engine returns tagged distances; ordered state updates preserve traversal, provided numerical results and tie rules also agree. Gathering alone would still transfer full vectors to host arithmetic. Local distance evaluation could avoid that payload, while query, descriptor, list and result transfers remain. The target needs local storage, suitable arithmetic, completion tracking and control over the modelled 32-byte request order. Whole-vector DMA or bank-local interfaces may not expose this cross-bank scheduling choice. This is

a design to test, not an assumed commercial-PIM interface or a new HNSW algorithm.

### A. Limits of the evidence

HNSW and public SIFT data permit stock-search tracing and exact subset checks at tractable simulation cost. They do not represent all ANN workloads. The evidence covers one corpus prefix and operating point: construction seeds do not test new datasets, dimensions, connectivity settings or recall targets. Those changes can alter both graph paths and the available candidate work. The smaller footprint tests transfer size while retaining float32 search; it does not implement quantisation. Ten paired splits estimate conditional query-selection variability, with no prospective power or universal sufficiency claim.

Flat replay knows future accesses early; bounded replay adds memory phases but approximates available overlap. Concurrent queries, caches, computation, scratch state, host/PIM transfers, energy, workload drift and actual mapping costs remain outside the evaluation. The simulator preset itself is not validated against a physical HBM2 device here.

## VII. Conclusion

The original contribution is a reproducible set of placement failure cases (C1), explained scheduling reversals (C2), and an execution proposal with cost thresholds (C3). The default 512-byte round-robin gain becomes a regression under the alternate mapper; its roughly 5,802 saved cycles per query also leave a finite allowance for extra cost. Designers can use these comparisons to test a gather-and-distance interface and reject unsupported scheduling assumptions. Exploratory profiling recovers useful fixed orders in this matrix. Application or PIM speedup still requires actual dependencies, computation, buffering, scheduling and communication to be measured.

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